Reference Verification Summary

This table checks the references against the CrossRef API to verify online existence, and extracts their exact citation contexts from the thesis.

Ref ID	Title	Exists Online?	Relevant?	Where Cited & Context
Ref1	2.5D Navigation Graph and Improved A-Star Algorithm for Path Planning in Ship inside Virtual Environment	Yes	Yes	**Chap2/chap2.tex**: `...g critical height variations and continuous surface characteristics \citep{Ref1, Ref2}. This coordinate abstraction allows pathfinding algorithms to e...`**Chap2/chap2.tex**: `...g critical height variations and continuous surface characteristics \citep{Ref1, Ref2}. This coordinate abstraction allows pathfinding algorithms to e...`**Chap2/chap2.tex**: `...a high-resolution Light Detection and Ranging (LiDAR) sensor suites \citep{Ref156}. To be utilized by global graph-search configurations, these unorga...`**Chap2/chap2.tex**: `...a high-resolution Light Detection and Ranging (LiDAR) sensor suites \citep{Ref156}. To be utilized by global graph-search configurations, these unorga...`**Chap2/chap2.tex**: `...a high-resolution Light Detection and Ranging (LiDAR) sensor suites \citep{Ref156}. To be utilized by global graph-search configurations, these unorga...`**Chap2/chap2.tex**: `...a high-resolution Light Detection and Ranging (LiDAR) sensor suites \citep{Ref156}. To be utilized by global graph-search configurations, these unorga...`**Chap2/chap2.tex**: `...in workspaces characterized by prominent vertical elevation changes \citep{Ref1, Ref228}. Furthermore, continuous parametric post-processing formulati...`
Ref2	Rapid Traversability Assessment in 2.5D Grid-based Map on Rough Terrain	Yes	Yes	**Chap2/chap2.tex**: `...ical height variations and continuous surface characteristics \citep{Ref1, Ref2}. This coordinate abstraction allows pathfinding algorithms to execute...`

Ref6	A 3D Positioning System for Off-Road Autonomous Vehicles	Maybe (Found: Environmental perception and m...)	Yes	**Chap2/chap2.tex**: `...NS) deliver high-frequency attitude rates and localized motion data \citep{Ref6}. The algorithmic integration of these asynchronous DGPS and INS measu...` **Chap2/chap2.tex**: `...NS) deliver high-frequency attitude rates and localized motion data \citep{Ref6}. The algorithmic integration of these asynchronous DGPS and INS measu...` **Chap2/chap2.tex**: `...NS) deliver high-frequency attitude rates and localized motion data \citep{Ref6}. The algorithmic integration of these asynchronous DGPS and INS measu...` **Chap2/chap2.tex**: `...NS) deliver high-frequency attitude rates and localized motion data \citep{Ref6}. The algorithmic integration of these asynchronous DGPS and INS measu...`
Ref10	A Comprehensive Review of Path Planning for Agricultural Ground Robots	No	Yes	**Chap2/chap2.tex**: `...lopments within advanced path planning and navigation architectures \citep{Ref10}. As established in Chapter 1, navigating complex topographies requir...` **Chap2/chap2.tex**: `...on seeding, automated harvesting, and broad-acre field surveillance \citep{Ref10, Ref14}. However, these unique field deployment scenarios impose rigo...` **Chap2/chap2.tex**: `...on seeding, automated harvesting, and broad-acre field surveillance \citep{Ref10, Ref14}. However, these unique field deployment scenarios impose rigo...` **Chap2/chap2.tex**: `...on seeding, automated harvesting, and broad-acre field surveillance \citep{Ref10, Ref14}. However, these unique field deployment scenarios impose rigo...` **Chap2/chap2.tex**: `...nmental terrain maps and tracking planned global trajectories \citep{Ref6, Ref10}....` **Chap2/chap2.tex**: `...e`

				understanding of terrain profiles to prevent platform instability \citep{Ref10}.... **Chap2/chap2.tex**: ...ng-endurance operations and economic viability in precision farming \citep{Ref10}. To model these thermal and fuel constraints without the complexity ...**Chap1/chap1.tex**: environments \citep{Ref10}. Unlike highly regulated urban streets or uniform industrial indoor**Chap1/chap1.tex**: over highly undulating landscapes \citep{Ref10}. To systematically address these multi-objective**Chap1/chap1.tex**: topologically aware understanding of terrain profiles to prevent platform instability \citep{Ref10}.**Chap1/chap1.tex**: 2.5D-informed cost functions \citep{Ref10, Ref560}. This empirical deficiency is particularly**Chap1/chap1.tex**: minimized cycle times and elevated productivity during field execution \citep{Ref10}. Operational**Chap1/chap1.tex**: battery depletion rates \citep{Ref10}. Additionally, the system's capacity to ingest static 2.5D
Ref14	A Multiple Mobile Robots Path Planning Algorithm Based on A-Star and Dijkstra Algorithm	Maybe (Found: Accessing the Best Path Using ...)	Yes	**Chap2/chap2.tex**: ...ing, automated harvesting, and broad-acre field surveillance \citep{Ref10, Ref14}. However, these unique field deployment scenarios impose rigorous an...
Ref29	A Path Planning Method for the Spherical Amphibious Robot Based on Improved A-Star Algorithm	Maybe (Found: An improved belief function me...)	Yes	**Chap2/chap2.tex**: ... vehicle from its starting configuration to the terminal goal state \citep{Ref29, Ref85, Ref309}. While conventional planar routines calculate this le...

Ref42	A Self-Supervised Near-to-Far Approach for Terrain-Adaptive Off-Road Autonomous Driving	Maybe (Found: RoadRunner—Learning Traversabi...)	Yes	**Chap2/chap2.tex**: ...`e's hard physical limits, such as its Maximum Climbable Slope (MCS) \citep{Ref42}. Trajectories optimized without these physical considerations may ap`...**Chap2/chap2.tex**: ...`e's hard physical limits, such as its Maximum Climbable Slope (MCS) \citep{Ref42}. Trajectories optimized without these physical considerations may ap`...**Chap1/chap1.tex**: `present severe vehicle control challenges \citep{Ref42}. Consequently, relying on flat surface`**Chap1/chap1.tex**: `drivability \citep{Ref96, Ref279, Ref42}. However, a significant deficiency exists in consistently`**Chap1/chap1.tex**: `platform, leading to unsafe or impossible execution profiles \citep{Ref42}. Currently, the explicit`
Ref45	A Survey of Path Planning Algorithms for Mobile Robots	Yes	Yes	**Chap2/chap2.tex**: ...`orithms operate effectively within low-overhead tracking boundaries \citep{Ref45}. However, this flat abstraction is fundamentally limited when deploy`...**Chap2/chap2.tex**: ...`al optimality and operational efficiency within static environments \citep{Ref45, Ref101}. Mechanistically, the algorithm systematically explores a gr`...**Chap2/chap2.tex**: ... `cost traversed from the initial start node to the current node $n$ \citep{Ref45, Ref101, Ref279}. Concurrently, $h(n)$ acts as a heuristic function e`...**Chap2/chap2.tex**: ... `cost traversed from the initial start node to the current node $n$ \citep{Ref45, Ref101, Ref279}. Concurrently, $h(n)$ acts as a heuristic function e`...**Chap2/chap2.tex**: ...`ch serves as a standard baseline across diverse pathfinding systems \citep{Ref45, Ref85, Ref279}.`

				Alternatively, when vehicle translations are constra...**Chap2/chap2.tex**: ...\noindent which is optimal for restrictive grid layouts \citep{Ref45}. For grid environments that explicitly accommodate uniform diagonal ...**Chap2/chap2.tex**: ...\noindent which minimizes structural distance distortion \citep{Ref45}. Selecting the most structurally appropriate heuristic from these op...**Chap2/chap2.tex**: ...arp, discontinuous waypoints that result in jagged or angular paths \citep{Ref45}. Executing these raw piece-wise linear path segments directly is hig...**Chap2/chap2.tex**: ...arp, discontinuous waypoints that result in jagged or angular paths \citep{Ref45}. Executing these raw piece-wise linear path segments directly is hig...**Chap2/chap2.tex**: ...o basic planar obstacle clearance and curvature smoothness criteria \citep{Ref45, Ref233}. However, robust off-road navigation architectures require t...**Chap2/chap2.tex**: ...ctuator wear, vehicle ride comfort, and localized mechanical stress \citep{Ref45, Ref45, Ref309, Ref507}. Integrating dedicated steering penalties or ...**Chap2/chap2.tex**: ...ctuator wear, vehicle ride comfort, and localized mechanical stress \citep{Ref45, Ref45, Ref309, Ref507}. Integrating dedicated steering penalties or ...**Chap2/chap2.tex**: ...ctuator wear, vehicle ride comfort, and localized mechanical stress \citep{Ref45, Ref45, Ref309, Ref507}. Integrating dedicated steering penalties or ...**Chap2/chap2.tex**: ...criteria tracking curvature smoothness and vehicle collision safety \citep{Ref45, Ref45,

				Ref54}....**Chap2/chap2.tex**: ...criteria tracking curvature smoothness and vehicle collision safety \citep{Ref45, Ref45, Ref54}....**Chap1/chap1.tex**: environments \citep{Ref45, Ref101}. These foundational frameworks operate by evaluating discrete**Chap1/chap1.tex**: similar spatial clearance metrics \citep{Ref45, Ref101}. For example, the deterministic search**Chap1/chap1.tex**: required to reach the terminal goal state \citep{Ref45, Ref101}. While these planar graph-search
Ref54	A Heuristic Approach for Solving the Online Capacitated Vehicle Routing Problem with Time Windows in Large-Scale Road Networks	Maybe (Found: A Novel Heuristic Algorithm fo...)	No (Manual Check Required)	**Chap2/chap2.tex**: ...d to complete graph exploration, cost updates, and validation loops \citep{Ref54, Ref108, Ref169, Ref309, Ref424}. Securing minimized runtime latencie...**Chap2/chap2.tex**: ...ing curvature smoothness and vehicle collision safety \citep{Ref45, Ref45, Ref54}....
Ref70	A Fast Two-Stage ACO Algorithm for Robotic Path Planning	Maybe (Found: Path Planning of Mobile Robot ...)	Yes	**Chap2/chap2.tex**: ...hronized data structures known as the Open List and the Closed List \citep{Ref70, Ref126, Ref144, Ref181}. The algorithm operates through iterative op...**Chap2/chap2.tex**: ...hronized data structures known as the Open List and the Closed List \citep{Ref70, Ref126, Ref144, Ref181}. The algorithm operates through iterative op...
Ref85	An Improved A-Star Algorithm Considering Water Current, Traffic Separation and Berthing for Vessel Path Planning	Maybe (Found: PID parameter tuning based on ...)	Yes	**Chap2/chap2.tex**: ...es as a standard baseline across diverse pathfinding systems \citep{Ref45, Ref85, Ref279}. Alternatively, when vehicle translations are constrained to...**Chap2/chap2.tex**: ..., traffic separation protocols, and localized maneuverability risks \citep{Ref85}. Similarly, off-road autonomous driving applications utilize modifie...**Chap2/chap2.tex**: ...e

				from its starting configuration to the terminal goal state \citep{Ref29, Ref85, Ref309}. While conventional planar routines calculate this length ov...**Chap2/chap2.tex**: ...odifiers—–such as fluid water currents for maritime vessel routing \citep{Ref85} or general collision and rollover boundary risks for off-road drivin...
Ref88	New Trends in Robotics for Agriculture: Integration and Assessment of a Real Fleet of Robots	Yes	Yes	**Chap2/chap2.tex**: ...nd an acute, long-term contraction of the skilled rural labor force \citep{Ref88}. To address these production bottlenecks, autonomous agricultural gr...**Chap1/chap1.tex**: labor pool \citep{Ref88}. To mitigate these macro-economic vulnerabilities, increasing the scale of**Chap1/chap1.tex**: \citep{Ref88}. Effective path planning algorithms designed for these agricultural platforms must
Ref96	AGV Path Planning Based on Improved A-Star Algorithm	Maybe (Found: Endoscopic Path Planning in Ro...)	Yes	**Chap1/chap1.tex**: drivability \citep{Ref96, Ref279, Ref42}. However, a significant deficiency exists in consistently
Ref101	A Method for the Shortest Path Search by Extended Dijkstra Algorithm	Maybe (Found: Parallelization of All-Pairs-S...)	Yes	**Chap2/chap2.tex**: ...mality and operational efficiency within static environments \citep{Ref45, Ref101}. Mechanistically, the algorithm systematically explores a graph or ...**Chap2/chap2.tex**: ...raversed from the initial start node to the current node n \citep{Ref45, Ref101, Ref279}. Concurrently, $h(n)$ acts as a heuristic function estimati...**Chap1/chap1.tex**: environments \citep{Ref45, Ref101}. These foundational frameworks operate by evaluating discrete**Chap1/chap1.tex**: similar spatial clearance metrics \citep{Ref45, Ref101}. For example, the deterministic search**Chap1/chap1.tex**: required to reach the terminal goal state

				\citep{Ref45, Ref101}. While these planar graph-search
Ref102	An Improved A-Star Path Planning Algorithm Based on Mobile Robots in Medical Testing Laboratories	Maybe (Found: Path Planning of Mobile Robot ...)	Yes	**Chap2/chap2.tex**: ...er while preserving the underlying topological features of the path \citep{Ref102, Ref482}....
Ref108	An Autonomous Navigation Approach for Unmanned Vehicle in Off-Road Environment with Self-Supervised Traversal Cost Prediction	Maybe (Found: Autonomous Navigation of KUVE ...)	Yes	**Chap2/chap2.tex**: ...mplete graph exploration, cost updates, and validation loops \citep{Ref54, Ref108, Ref169, Ref309, Ref424}. Securing minimized runtime latencies is pa...**Chap2/chap2.tex**: ...ering structural collisions or cross-country immobilization hazards \citep{Ref108, Ref253, Ref309}. In non-planar environments, this validation parame...**Chap2/chap2.tex**: ...ticularly when confronting complex and hazardous off-road scenarios \citep{Ref108, Ref134, Ref154}. High-fidelity virtual simulations facilitate compl...**Chap2/chap2.tex**: ...ticularly when confronting complex and hazardous off-road scenarios \citep{Ref108, Ref134, Ref154}. High-fidelity virtual simulations facilitate compl...
Ref126	Application SLAM and Path Planning using A-Star Algorithm for Mobile Robot in Indoor Disaster Area	Maybe (Found: Disaster Recovery Planning...)	Yes	**Chap2/chap2.tex**: ...d data structures known as the Open List and the Closed List \citep{Ref70, Ref126, Ref144, Ref181}. The algorithm operates through iterative optimizat...**Chap2/chap2.tex**: ...d data structures known as the Open List and the Closed List \citep{Ref70, Ref126, Ref144, Ref181}. The algorithm operates through iterative optimizat...
Ref134	Autonomous Off-Road Navigation	Maybe (Found: Terrain	Yes	**Chap2/chap2.tex**: ...y when confronting complex and hazardous off-road scenarios \citep{Ref108,

	Using Stereo-Vision and Laser-Rangefinder Fusion for Outdoor Obstacles Detection	mapping for off-road A...)		Ref134, Ref154}. High-fidelity virtual simulations facilitate completely re...
Ref144	Development of Path Planning Approach Using Improved A-Star Algorithm in AGV System	Maybe (Found: Optimising Multi-AGV Path Plan...)	Yes	**Chap2/chap2.tex**: `...tructures known as the Open List and the Closed List \citep{Ref70, Ref126, Ref144, Ref181}. The algorithm operates through iterative optimization loop...`
Ref152	Enhanced Hybrid Path Planning Algorithm Based on APF and A-Star	No	Yes	**Chap2/chap2.tex**: `...selection loops to guide the vehicle out of dead-end configurations \citep{Ref152}. These reactive and force-based local planners provide the essentia...`
Ref154	Multisensor Data Fusion for Robust Pose Estimation of a Six-Legged Walking Robot	Yes	Yes	**Chap2/chap2.tex**: `...onfronting complex and hazardous off-road scenarios \citep{Ref108, Ref134, Ref154}. High-fidelity virtual simulations facilitate completely reproducib...`
Ref156	Enabling Off-Road Autonomous Navigation---Simulation of LIDAR in Dense Vegetation	Maybe (Found: Simulating Off-Road Autonomous...)	Yes	**Chap2/chap2.tex**: `...a high-resolution Light Detection and Ranging (LiDAR) sensor suites \citep{Ref156}. To be utilized by global graph-search configurations, these unorga...` **Chap2/chap2.tex**: `...ntial locomotion resistance or risk inducing structural crop damage \citep{Ref156}. These non-binary cost layers operate by dynamically injecting scal...`
Ref169	A Survey of Autonomous Mobile Robot Path Planning Approaches	Maybe (Found: Combining proactive planning a...)	Yes	**Chap2/chap2.tex**: `...raph exploration, cost updates, and validation loops \citep{Ref54, Ref108, Ref169, Ref309, Ref424}. Securing minimized runtime latencies is paramount ...` **Chap1/chap1.tex**: `dynamic obstacles \citep{Ref169, Ref199, Ref396}, they fail to`

				systematically integrate multi-**Chap1/chap1.tex**: obstacle avoidance, and navigate transient dynamic environments \citep{Ref169, Ref199, Ref396},
Ref181	A Real-time Collision Avoidance Algorithm for Autonomous Surface Vessels	Maybe (Found: Adjacent Cross-Coupling Synchr...)	Yes	**Chap2/chap2.tex**: ...s known as the Open List and the Closed List \citep{Ref70, Ref126, Ref144, Ref181}. The algorithm operates through iterative optimization loops, conti...
Ref199	Global Path Planning of Unmanned Ground Vehicles in Large-Scale Off-Road Environments Based on Improved A* Algorithm and Quadratic Programming	Maybe (Found: An improved SUSAN corner detec...)	Yes	**Chap1/chap1.tex**: dynamic obstacles \citep{Ref169, Ref199, Ref396}, they fail to systematically integrate multi-**Chap1/chap1.tex**: obstacle avoidance, and navigate transient dynamic environments \citep{Ref169, Ref199, Ref396},
Ref206	Improved Hybrid A-Star Algorithm for Path Planning in Autonomous Parking System Based on Multi-Stage Dynamic Optimization	Yes	Yes	**Chap2/chap2.tex**: ...primitives to guarantee drivable trajectories in constrained spaces \citep{Ref206}. While Hybrid A* effectively secures kinematic feasibility, its sta...
Ref210	Improved Hierarchical A-Star Algorithm for Optimal Parking Path Planning of the Large Parking Lot	Yes	Yes	**Chap2/chap2.tex**: ...romise operational safety, or accelerate vehicle energy consumption \citep{Ref210, Ref309, Ref424}. For instance, in maritime routing domains, the cor...**Chap2/chap2.tex**: ...hes frequently treat vehicle speed as a static, unvarying parameter \citep{Ref210}, advanced field robotics frameworks incorporate dynamic speed-reduc...

Ref212	Guided Hybrid A-Star Path Planning Algorithm for Valet Parking Applications	Yes	Yes	**Chap1/chap1.tex**: `architectures represents an indispensable algorithmic prerequisite \citep{Ref212}. Consequently,`
Ref214	Integrated Air/Ground Vehicle System for Semi-Autonomous Off-Road Navigation	Maybe (Found: Autonomous Vehicle...)	Yes	**Chap2/chap2.tex**: `...fety hazards associated with real-world mobile platform deployments \citep{Ref214}. Open-source robotics middleware and simulators, most notably Gazeb...`
Ref216	Three Engineers, Hundreds of Robots, One Warehouse	Maybe (Found: These Humanoid Robots Could Ki...)	Yes	**Chap1/chap1.tex**: `productivity across diverse sectors \citep{Ref216}. Within this broader context, off-road autonomous`
Ref228	Indoor 2.5D Navigation for Ground Robots	Yes	Yes	**Chap2/chap2.tex**: `...environment that completely neglects the crucial vertical dimension \citep{Ref228}. This structural oversight routinely leads to the generation of glo...`**Chap2/chap2.tex**: `...environment that completely neglects the crucial vertical dimension \citep{Ref228}. This structural oversight routinely leads to the generation of glo...`**Chap2/chap2.tex**: `...environment that completely neglects the crucial vertical dimension \citep{Ref228}. This structural oversight routinely leads to the generation of glo...`**Chap2/chap2.tex**: `...le paths, structural collisions, and catastrophic navigation errors \citep{Ref228}. Consequently, relying on flat surface assumptions strips the navig...`**Chap2/chap2.tex**: `...g traversability and calculating instantaneous movement costs \citep{Ref1, Ref228}. By integrating these vital vertical dimensions without inheriting ...`**Chap2/chap2.tex**: `...kspaces`

				characterized by prominent vertical elevation changes \citep{Ref1, Ref228}. Furthermore, continuous parametric post-processing formulations, s...**Chap1/chap1.tex**: elevation dimension \citep{Ref228}. This geometric omission can lead to severe navigation failures,**Chap1/chap1.tex**: untraversable due to excessive, unsafe slope gradients \citep{Ref228}. To resolve these spatial**Chap1/chap1.tex**: system to derive key topographic features like slope gradients \citep{Ref228}. This strategic**Chap1/chap1.tex**: densities associated with full 3D volumetric representations \citep{Ref228}. Transitioning to these**Chap1/chap1.tex**: dimensional (2D) path planning paradigms \citep{Ref228}. Relying exclusively on 2D planar**Chap1/chap1.tex**: large elevation changes, or undetected vertical obstacles \citep{Ref228}. For instance, a planar**Chap1/chap1.tex**: navigation errors \citep{Ref228}. Furthermore, attempting to retroactively fix these geometric**Chap1/chap1.tex**: and specific operational challenges encountered by agricultural robots \citep{Ref228}. Consequently,
Ref233	Local Off-Road Navigation	No	Yes	**Chap2/chap2.tex**: ...h geometric dimensionality that incur severe computational overhead \citep{Ref233}. This processing burden demands substantial runtime memory storage ...**Chap2/chap2.tex**: ...h geometric dimensionality that incur severe computational overhead \citep{Ref233}. This processing burden demands substantial runtime memory storage ...**Chap2/chap2.tex**: ... planar obstacle clearance and curvature smoothness criteria \citep{Ref45, Ref233}. However, robust off-road navigation architectures require the inte...**Chap1/chap1.tex**:

				\citep{Ref233}. While widespread modifications to the classical A* graph-search algorithm
Ref253	Learning-based Uncertainty-aware Navigation in 3D Off-Road Terrains	Yes	Yes	**Chap2/chap2.tex**: ...ructural collisions or cross-country immobilization hazards \citep{Ref108, Ref253, Ref309}. In non-planar environments, this validation parameter focu...
Ref266	Methodology for Path Planning and Optimization of Mobile Robots: A Review	Yes	Yes	**Chap2/chap2.tex**: ...on of personal historical optima and global swarm best-found states \citep{Ref266}. In agricultural routing, PSO frameworks have been implemented to g...**Chap2/chap2.tex**: ...otential field that drives the vehicle away from structural hazards \citep{Ref266, Ref491}. While APF formulations are computationally inexpensive and...**Chap2/chap2.tex**: ...otential field that drives the vehicle away from structural hazards \citep{Ref266, Ref491}. While APF formulations are computationally inexpensive and...
Ref277	Improving Robot Navigation Through Self-Supervised Online Learning	Yes	Yes	**Chap2/chap2.tex**: ...ion systems compared to highly regulated, structured urban settings \citep{Ref277}. These unstructured spaces are characterized by unpredictable macro...**Chap2/chap2.tex**: ...ion systems compared to highly regulated, structured urban settings \citep{Ref277}. These unstructured spaces are characterized by unpredictable macro...**Chap2/chap2.tex**: ...ion systems compared to highly regulated, structured urban settings \citep{Ref277}. These unstructured spaces are characterized by unpredictable macro...**Chap1/chap1.tex**: obstacles \citep{Ref277}. To guarantee reliable vehicle state estimation and successful terrain

Ref279	Mobile Robot Path Planning Algorithm Based on Improved A Star	Yes	Yes	**Chap2/chap2.tex**: ... from the initial start node to the current node n \citep{Ref45, Ref101, Ref279}. Concurrently, $h(n)$ acts as a heuristic function estimating the r...**Chap2/chap2.tex**: ... standard baseline across diverse pathfinding systems \citep{Ref45, Ref85, Ref279}. Alternatively, when vehicle translations are constrained to standa...**Chap2/chap2.tex**: ...tracking instabilities within low-level feedback controllers \citep{Ref45, Ref279, Ref470}. Therefore, implementing an algorithmic post-processing ref...**Chap2/chap2.tex**: ...erty and their ability to generate continuous, differentiable paths \citep{Ref279}. This mathematical formulation allows engineers to alter local curv...**Chap2/chap2.tex**: ...erty and their ability to generate continuous, differentiable paths \citep{Ref279}. This mathematical formulation allows engineers to alter local curv...**Chap2/chap2.tex**: ...aypoints into kinematically executable curvilinear functions \citep{Ref45, Ref279}....**Chap2/chap2.tex**: ...nto smooth, differentiable, and kinematically suitable trajectories \citep{Ref279}. To quantify the efficacy of these integrated navigation systems, s...**Chap1/chap1.tex**: drivability \citep{Ref96, Ref279, Ref42}. However, a significant deficiency exists in consistently
Ref283	Mobile Robot Path Planning Based on Kinematically Constrained A-Star Algorithm and DWA	Yes	Yes	**Chap2/chap2.tex**: ...et headings, obstacle clearance distances, and localized velocities \citep{Ref283, Ref396, Ref482}. By selecting the velocity command that maximizes t...

	Fusion Algorithm			
Ref284	Mobile Robot Path Planning Based on Improved Genetic Algorithm with A-Star Heuristic Method	Yes	Yes	**Chap2/chap2.tex**: ...gh-quality heuristic profiles and accelerate structural convergence \citep{Ref284}. Similarly, Particle Swarm Optimization (PSO) maps candidate soluti...
Ref309	Multi-objective Multi-robot Path Planning in Continuous Environment Using an Enhanced Genetic Algorithm	Yes	Yes	**Chap2/chap2.tex**: ...perational safety, or accelerate vehicle energy consumption \citep{Ref210, Ref309, Ref424}. For instance, in maritime routing domains, the core A* arc...**Chap2/chap2.tex**: ...eing high-order parametric continuity across the trajectory profile \citep{Ref309}. Specifically, enforcing C^2 continuity---which mathematically se...**Chap2/chap2.tex**: ...its starting configuration to the terminal goal state \citep{Ref29, Ref85, Ref309}. While conventional planar routines calculate this length over a fl...**Chap2/chap2.tex**: ...its starting configuration to the terminal goal state \citep{Ref29, Ref85, Ref309}. While conventional planar routines calculate this length over a fl...**Chap2/chap2.tex**: ...its starting configuration to the terminal goal state \citep{Ref29, Ref85, Ref309}. While conventional planar routines calculate this length over a fl...**Chap2/chap2.tex**: ... collisions or cross-country immobilization hazards \citep{Ref108, Ref253, Ref309}. In non-planar environments, this validation parameter focuses on a...
Ref368	Optimal Path Planning for	No	Yes	**Chap2/chap2.tex**: ...ectively identify optimal routes via

	Drones Based on Swarm Intelligence Algorithm			simulated pheromone deposition \citep{Ref368, Ref495}. Although ACO architectures require substantial computation...**Chap2/chap2.tex**: ...ectively identify optimal routes via simulated pheromone deposition \citep{Ref368, Ref495}. Although ACO architectures require substantial computation...**Chap1/chap1.tex**: field machinery and prevent localized crop damage \citep{Ref368}. Concurrently, optimizing paths
Ref380	PQ-RRT*: An Improved Path Planning Algorithm for Mobile Robots	Yes	Yes	**Chap2/chap2.tex**: ...ces, offering mathematical guarantees of probabilistic completeness \citep{Ref380, Ref406}. Mechanistically, the standard RRT algorithm explores the v...**Chap2/chap2.tex**: ...ces, offering mathematical guarantees of probabilistic completeness \citep{Ref380, Ref406}. Mechanistically, the standard RRT algorithm explores the v...
Ref396	Path Planning Based on Combination of Improved A-STAR Algorithm and DWA Algorithm	Yes	Yes	**Chap2/chap2.tex**: ...ngs, obstacle clearance distances, and localized velocities \citep{Ref283, Ref396, Ref482}. By selecting the velocity command that maximizes this reac...**Chap1/chap1.tex**: dynamic obstacles \citep{Ref169, Ref199, Ref396}, they fail to systematically integrate multi-**Chap1/chap1.tex**: obstacle avoidance, and navigate transient dynamic environments \citep{Ref169, Ref199, Ref396},
Ref406	Path Planning for Autonomous Mobile Robots: A Review	Yes	Yes	**Chap2/chap2.tex**: ...ering mathematical guarantees of probabilistic completeness \citep{Ref380, Ref406}. Mechanistically, the standard RRT algorithm explores the vehicle's...**Chap2/chap2.tex**: ...ering mathematical guarantees of probabilistic completeness \citep{Ref380, Ref406}. Mechanistically, the standard RRT

				algorithm explores the vehicle's...**Chap2/chap2.tex**: ...ering mathematical guarantees of probabilistic completeness \citep{Ref380, Ref406}. Mechanistically, the standard RRT algorithm explores the vehicle's...**Chap1/chap1.tex**: constraints and avoiding localized environmental obstacles \citep{Ref406}. Historically, classical
Ref410	Path Planning for Virtual Human Motion Using Improved A* Algorithm	Yes	Yes	**Chap2/chap2.tex**: ...erestimates the true remaining cost---and mathematically consistent \citep{Ref410}. This core cost infrastructure underpins the node selection strateg...**Chap2/chap2.tex**: ...shapes the algorithm's runtime efficiency and exploration footprint \citep{Ref410}....**Chap2/chap2.tex**: ...arch exploration ($w < 1$) and focused heuristic guidance ($w > 1$) \citep{Ref410}. Building upon this weighting concept, the framework developed in t...
Ref417	Path Planning for the Mobile Robot: A Review	Yes	Yes	**Chap2/chap2.tex**: ...roviding robust global search capabilities across non-linear spaces \citep{Ref417, Ref519}. These evolutionary mechanics can be combined with A* cost ...
Ref424	Path Planning with Modified A* Algorithm for a Mobile Robot	Maybe (Found: Deep Reinforcement Learning Al...)	Yes	**Chap2/chap2.tex**: ...al safety, or accelerate vehicle energy consumption \citep{Ref210, Ref309, Ref424}. For instance, in maritime routing domains, the core A* architectur...**Chap2/chap2.tex**: ..., cost updates, and validation loops \citep{Ref54, Ref108, Ref169, Ref309, Ref424}. Securing minimized runtime latencies is paramount for ensuring edg...**Chap2/chap2.tex**: ...tive energy consumption proxies (ECP) within their evaluation loops \citep{Ref424}. These energy proxies

				operate by applying non-linear penalties to p...
Ref440	Path Planning of Automated Guided Vehicles Based on Improved A-Star Algorithm	Yes	Yes	**Chap2/chap2.tex**: `...pe inclinations, surface roughness, or localized obstacle proximity \citep{Ref440}. Concurrently, optimizing mechanical energy expenditure is a core r...`
Ref452	Path Planning Techniques for Mobile Robots: Review and Prospect	Yes	Yes	**Chap2/chap2.tex**: `...rom the nearest existing vertex toward these target samples \citep{Ref380, Ref452}. Concurrently, the PRM architecture constructs a continuous topolog...`
Ref470	Precise Path Planning and Trajectory Tracking Based on Improved A-Star Algorithm	Yes	Yes	**Chap2/chap2.tex**: `... instabilities within low-level feedback controllers \citep{Ref45, Ref279, Ref470}. Therefore, implementing an algorithmic post-processing refinement ...`
Ref471	Real-Time Parametric Path Planning Algorithm for Agricultural Machinery Kinematics Model Based on Particle Swarm Optimization	Yes	Yes	**Chap2/chap2.tex**: `...e vehicle kinematic configurations alongside static terrain hazards \citep{Ref471}. Furthermore, Ant Colony Optimization (ACO) mimics swarm intelligen...`
Ref482	Research on Path Planning with the Integration of Adaptive A-Star Algorithm and Improved Dynamic Window Approach	Yes	Yes	**Chap2/chap2.tex**: `...tacle clearance distances, and localized velocities \citep{Ref283, Ref396, Ref482}. By selecting the velocity command that maximizes this reactive fun...`**Chap2/chap2.tex**: `... preserving the underlying topological features of the path \citep{Ref102, Ref482}....`
Ref491	Research on Path-Planning Algorithm Integrating Optimization	Yes	Yes	**Chap2/chap2.tex**: `... field that drives the vehicle away from structural hazards \citep{Ref266, Ref491}. While APF formulations are computationally inexpensive and`

	A-Star Algorithm and Artificial Potential Field Method			highly ...**Chap2/chap2.tex**: ... field that drives the vehicle away from structural hazards \citep{Ref266, Ref491}. While APF formulations are computationally inexpensive and highly ...
Ref495	Research on Path Planning of Mobile Robot Based on Improved Ant Colony Algorithm	Yes	Yes	**Chap2/chap2.tex**: ... identify optimal routes via simulated pheromone deposition \citep{Ref368, Ref495}. Although ACO architectures require substantial computational itera...
Ref507	Rollover-Free Path Planning for Off-Road Autonomous Driving	Yes	Yes	**Chap2/chap2.tex**: ...terrains to actively prevent catastrophic rollover accidents \citep{Ref10, Ref507}. Consequently, achieving high operational precision during in-field...**Chap2/chap2.tex**: ...n indices and rollover risks across dynamic environmental grid maps \citep{Ref507}. However, despite these advancements, the direct, real-time integra...**Chap2/chap2.tex**: ...path selection to guarantee platform stability over severe inclines \citep{Ref507}....**Chap2/chap2.tex**: ...ride comfort, and localized mechanical stress \citep{Ref45, Ref45, Ref309, Ref507}. Integrating dedicated steering penalties or post-processing B-spli...**Chap2/chap2.tex**: ... general collision and rollover boundary risks for off-road driving \citep{Ref507}---the literature exhibits a clear absence of a unified, explicit co...**Chap1/chap1.tex**: to catastrophic rollovers, collisions, or immobilization on untraversable gradients \citep{Ref507}.
Ref519	Survey on Computational-Intelligence-Based UAV Path Planning	Yes	Yes	**Chap2/chap2.tex**: ... robust global search capabilities across non-linear spaces \citep{Ref417, Ref519}. These evolutionary

				mechanics can be combined with A* cost models t...
Ref560	Smoothed A* Algorithm for Practical Unmanned Surface Vehicle Path Planning	Yes	Yes	**Chap1/chap1.tex**: `2.5D-informed cost functions \citep{Ref10, Ref560}. This empirical deficiency is particularly`